Amazon
AIF-C01
AWS Certified AI Practitioner Exam

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Question: 1

An AI practitioner trained a custom model on Amazon Bedrock by using a training dataset that contains confidential data. The AI practitioner wants to ensure that the custom model does not generate inference responses based on confidential data.
How should the AI practitioner prevent responses based on confidential data?
A. Delete the custom model. Remove the confidential data from the training dataset. Retrain the custom model.
B. Mask the confidential data in the inference responses by using dynamic data masking.
C. Encrypt the confidential data in the inference responses by using Amazon SageMaker.
D. Encrypt the confidential data in the custom model by using AWS Key Management Service (AWS KMS).

Answer: A

Explanation:
When a model is trained on a dataset containing confidential or sensitive data, the model may inadvertently learn patterns from this data, which could then be reflected in its inference responses. To ensure that a model does not generate responses based on confidential data, the most effective approach is to remove the confidential data from the training dataset and then retrain the model.
Explanation of Each Option:
Option A (Correct): "Delete the custom model. Remove the confidential data from the training dataset. Retrain the custom model."This option is correct because it directly addresses the core issue: the model has been trained on confidential data. The only way to ensure that the model does not produce inferences based on this data is to remove the confidential information from the training dataset and then retrain the model from scratch. Simply deleting the model and retraining it ensures that no confidential data is learned or retained by the model. This approach follows the best practices recommended by AWS for handling sensitive data when using machine learning services like Amazon Bedrock.
Option B: "Mask the confidential data in the inference responses by using dynamic data masking."This option is incorrect because dynamic data masking is typically used to mask or obfuscate sensitive data in a database. It does not address the core problem of the model beingtrained on confidential data. Masking data in inference responses does not prevent the model from using confidential data it learned during training.
Option C: "Encrypt the confidential data in the inference responses by using Amazon SageMaker."This option is incorrect because encrypting the inference responses does not prevent the model from generating outputs based on confidential data. Encryption only secures the data at rest or in transit but does not affect the model's underlying knowledge or training process.
Option D: "Encrypt the confidential data in the custom model by using AWS Key Management Service (AWS KMS)."This option is incorrect as well because encrypting the data within the model does not prevent the model from generating responses based on the confidential data it learned during training. AWS KMS can encrypt data, but it does not modify the learning that the model has already performed.

AWS AI Practitioner Reference:
Data Handling Best Practices in AWS Machine Learning: AWS advises practitioners to carefully handle training data, especially when it involves sensitive or confidential information. This includes preprocessing steps like data anonymization or removal of sensitive data before using it to train machine learning models.
Amazon Bedrock and Model Training Security: Amazon Bedrock provides foundational models and customization capabilities, but any training involving sensitive data should follow best practices, such as removing or anonymizing confidential data to prevent unintended data leakage.

Question: 2

Which feature of Amazon OpenSearch Service gives companies the ability to build vector database applications?

A. Integration with Amazon S3 for object storage
B. Support for geospatial indexing and queries
C. Scalable index management and nearest neighbor search capability
D. Ability to perform real-time analysis on streaming data

Answer: C

Explanation:
Amazon OpenSearch Service (formerly Amazon Elasticsearch Service) has introduced capabilities to support vector search, which allows companies to build vector database applications. This is particularly useful in machine learning, where vector representations (embeddings) of data are often used to capture semantic meaning.
Scalable index management and nearest neighbor search capability are the core features enabling vector database functionalities in OpenSearch. The service allows users to index high-dimensional vectors and perform efficient nearest neighbor searches, which are crucial for tasks such as recommendation systems, anomaly detection, and semantic search.
Here is why option C is the correct answer:
Scalable Index Management: OpenSearch Service supports scalable indexing of vector data. This means you can index a large volume of high-dimensional vectors and manage these indexes in a cost-effective and performance-optimized way. The service leverages underlying AWS infrastructure to ensure that indexing scales seamlessly with data size.
Nearest Neighbor Search Capability: OpenSearch Service's nearest neighbor search capability allows for fast and efficient searches over vector data. This is essential for applications like product recommendation engines, where the system needs to quickly find the most similar items based on a user's query or behavior.
AWS AI Practitioner Reference:
According to AWS documentation, OpenSearch Service's support for nearest neighbor search using vector embeddings is a key feature for companies building machine learning applications that require similarity search.
The service uses Approximate Nearest Neighbors (ANN) algorithms to speed up searches over large datasets, ensuring high performance even with large-scale vector data.
The other options do not directly relate to building vector database applications:

A . Integration with Amazon S3 for object storage is about storing data objects, not vector-based searching or indexing.
B . Support for geospatial indexing and queries is related to location-based data, not vectors used in machine learning.
D . Ability to perform real-time analysis on streaming data relates to analyzing incoming data streams, which is different from the vector search capabilities.

Question: 3

A company wants to display the total sales for its top-selling products across various retail locations in the past 12 months.
Which AWS solution should the company use to automate the generation of graphs?

A. Amazon Q in Amazon EC2
B. Amazon Q Developer
C. Amazon Q in Amazon QuickSight
D. Amazon Q in AWS Chatbot

Answer: C

Explanation:
Amazon QuickSight is a fully managed business intelligence (BI) service that allows users to create and publish interactive dashboards that include visualizations like graphs, charts, and tables. "Amazon Q" is the natural language query feature within Amazon QuickSight. It enables users to ask questions about their data in natural language and receive visual responses such as graphs.
Option C (Correct): "Amazon Q in Amazon QuickSight": This is the correct answer because Amazon QuickSight Q is specifically designed to allow users to explore their data through natural language queries, and it can automatically generate graphs to display sales data and other metrics. This makes it an ideal choice for the company to automate the generation of graphs showing total sales for its topselling products across various retail locations.
Option A, B, and D: These options are incorrect:
A . Amazon Q in Amazon EC2: Amazon EC2 is a compute service that provides virtual servers, but it is not directly related to generating graphs or providing natural language querying features.
B . Amazon Q Developer: This is not an existing AWS service or feature.
D . Amazon Q in AWS Chatbot: AWS Chatbot is a service that integrates with Amazon Chime and Slack for monitoring and managing AWS resources, but it is not used for generating graphs based on sales data.
AWS AI Practitioner Reference:
Amazon QuickSight Q is designed to provide insights from data by using natural language queries, making it a powerful tool for generating automated graphs and visualizations directly from queried data.
Business Intelligence (BI) on AWS: AWS services such as Amazon QuickSight provide business intelligence capabilities, including automated reporting and visualization features, which are ideal for companies seeking to visualize data like sales trends over time.

Question: 4

A company wants to build an interactive application for children that generates new stories based on classic stories. The company wants to use Amazon Bedrock and needs to ensure that the results and topics are appropriate for children.
Which AWS service or feature will meet these requirements?

A. Amazon Rekognition
B. Amazon Bedrock playgrounds
C. Guardrails for Amazon Bedrock
D. Agents for Amazon Bedrock

Answer: C

Explanation:
Amazon Bedrock is a service that provides foundational models for building generative AI applications. When creating an application for children, it is crucial to ensure that the generated content is appropriate for the target audience. "Guardrails" in Amazon Bedrock provide mechanisms to control the outputs and topics of generated content to align with desired safety standards and appropriateness levels.
Option C (Correct): "Guardrails for Amazon Bedrock": This is the correct answer because guardrails are specifically designed to help users enforce content moderation, filtering, and safety checks on the outputs generated by models in Amazon Bedrock. For a children's application, guardrails ensure that all content generated is suitable and appropriate for the intended audience.
Option A: "Amazon Rekognition" is incorrect. Amazon Rekognition is an image and video analysis service that can detect inappropriate content in images or videos, but it does not handle text or story generation.
Option B: "Amazon Bedrock playgrounds" is incorrect because playgrounds are environments for experimenting and testing model outputs, but they do not inherently provide safeguards to ensure content appropriateness for specific audiences, such as children.
Option D: "Agents for Amazon Bedrock" is incorrect. Agents in Amazon Bedrock facilitate building AI applications with more interactive capabilities, but they do not provide specific guardrails for ensuring content appropriateness for children.
AWS AI Practitioner Reference:
Guardrails in Amazon Bedrock: Designed to help implement controls that ensure generated content is safe and suitable for specific use cases or audiences, such as children, by moderating and filtering inappropriate or undesired content.
Building Safe AI Applications: AWS provides guidance on implementing ethical AI practices, including using guardrails to protect against generating inappropriate or biased content.

Question: 5

A company has developed an ML model for image classification. The company wants to deploy the model to production so that a web application can use the model.

The company needs to implement a solution to host the model and serve predictions without managing any of the underlying infrastructure.
Which solution will meet these requirements?

A. Use Amazon SageMaker Serverless Inference to deploy the model.
B. Use Amazon CloudFront to deploy the model.
C. Use Amazon API Gateway to host the model and serve predictions.
D. Use AWS Batch to host the model and serve predictions.

Answer: A

Explanation:
Amazon SageMaker Serverless Inference is the correct solution for deploying an ML model to production in a way that allows a web application to use the model without the need to manage the underlying infrastructure.
Amazon SageMaker Serverless Inference provides a fully managed environment for deploying machine learning models. It automatically provisions, scales, and manages the infrastructure required to host the model, removing the need for the company to manage servers or other underlying infrastructure.
Why Option A is Correct:
No Infrastructure Management: SageMaker Serverless Inference handles the infrastructure management
for deploying and serving ML models. The company can simply provide the model and specify the required compute capacity, and SageMaker will handle the rest.
Cost-Effectiveness: The serverless inference option is ideal for applications with intermittent or unpredictable traffic, as the company only pays for the compute time consumed while handling requests.
Integration with Web Applications: This solution allows the model to be easily accessed by web applications via RESTful APIs, making it an ideal choice for hosting the model and serving predictions.
Why Other Options are Incorrect:
B . Use Amazon CloudFront to deploy the model: CloudFront is a content delivery network (CDN) service for distributing content, not for deploying ML models or serving predictions.
C . Use Amazon API Gateway to host the model and serve predictions: API Gateway is used for creating, deploying, and managing APIs, but it does not provide the infrastructure or the required environment to host and run ML models.
D . Use AWS Batch to host the model and serve predictions: AWS Batch is designed for running batch computing workloads and is not optimized for real-time inference or hosting machine learning models.
Thus, A is the correct answer, as it aligns with the requirement of deploying an ML model without managing any underlying infrastructure.

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